

Tuan-Anh VU

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RESEARCH GOAL

Develop learning-enabled robotic systems that achieve **robust perception**, **actionable 3D/4D world modeling**, and **safe autonomy**—unifying *perception-cognition-action* and validating performance on real platforms through rigorous, repeatable experimentation. My long-term goal is to build **trustworthy embodied intelligence** for real-world autonomy: robots that (i) perceive and reason under distribution shift and visual ambiguity, (ii) maintain accurate **3D/4D** representations of changing scenes, and (iii) safely execute autonomy for navigation, monitoring/inspection, and targeted intervention.

PROFILE HIGHLIGHTS

- **Research and collaboration:** Current research lies on the intersection of **Computer Vision, Robotics, and Physics-Guided AI**, with a focus on creating resilient AI systems capable of joint object reconstruction and motion estimation in 4D space, robust scene understanding, specifically addressing occlusions and complex environmental cues.
- **Publications and awards:** 16 peer-reviewed publications and 20 manuscripts under review or in preprint, including 13 first- or co-first-authored works within 6 years in top journals and conferences. My work has been recognized with numerous **Oral main conference or workshop presentations** at flagship venues including ECCV, CVPR, AAAI, ICCP, and MMM. Received **International Recognitions** (ARAP Research Award 2023-2024 of approximately 80,000 USD, Hong Kong UGC Research Travel Grant 2022 & 2024, EPFL CIS Edge AI Summer School Travel Grant 2022, and Best Poster Award at Machine Learning Summer School series in Indonesia 2020), **Doctoral Consortia** (IEEE/CVF WACV 2025, ECCV 2024, and IEEE CAI 2024), and **Competitive Challenges** (2nd Place at USV-based Obstacle Segmentation Challenge in two consecutive years 2023 & 2024 at the WACV Workshop on Maritime Computer Vision).
- **Grant Writing:** Named **Senior Personnel** on a \$1M National Science Foundation (NSF) CPS award, having generated the preliminary data and contributed to the proposal. Served as PI/Co-PI on more than \$2M in high-performance computing and cloud allocations.
- **Mentorship and Teaching:** Mentored 2 Ph.D. students, 3 master's students, and 11 undergraduates. Served as a teaching assistant for more than 15 undergraduate and postgraduate course offerings in computer science, programming, multimedia computing, digital design, and artificial intelligence.
- **Leadership and Professional Service:** CEO of Computer Vision for Developing Countries (NPO); Executive Officer-Secretary of the IEEE Coastal Los Angeles Section; organizer of the CV4DC and CV4Animals workshops; and active scholarly contributor as a guest editor, area chair, session co-chair, and reviewer for leading conferences and journals.

EDUCATION

- 2019 - 2024 **Ph.D. in Computer Science and Engineering**, The Hong Kong University of Science and Technology
Supervisor: Prof. Sai-Kit Yeung
Dissertation title: Robust Scene Understanding in Challenging Scenarios
- 2011 - 2016 **B.Sc. in Computer Science**, International University - Vietnam National University HCMC (HCMIU-VNU)
Supervisor: Senior Lecturer Synh Viet-Uyen Ha
Thesis title: Extend Traffic Signs Detection and Recognition Algorithm in Nighttime in Viet Nam

GRANT & FUNDING

- AUG 2026 **NSF CPS-CIR, CPS-FR – Senior Personnel** (PI: M. K. Jawed); \$1,000,000 total award for "Physics-Guided Latent Space Models for Detecting Occluded Objects" ([NSF Award 2551220](#))
- AUG 2026 **NSF ACCESS Accelerate (Co-PI)**, received 1,500,000 credits for using high-performance computing, and cloud services ([ACCESS allocation CIS261206](#))
- AUG 2026 **Thinking Machines Lab Tinker Research Grant (PI)**, received \$5,000 credits for using cloud computing
- MAY 2026 **NSF ACCESS Discover (PI)**, received 750,000 credits for using high-performance computing, and cloud services ([ACCESS allocation CIS260951](#))
- MAY 2026 **NAIRR Pilot Start-Up (PI)**, received 2,000 GPU hours for using high-performance computing ([NAIRR Pilot NAIRR260204](#))

HONORS & AWARDS

- JUL 2026 **Outstanding Reviewer**, ECCV'26 (782 / 12,280 $\approx$ 6.37% - [Link](#))
- JAN 2025 **Doctoral Consortium (Travel Award)**, WACV'25, mentored by Prof. Richard Souvenir and Sharon X. Huang
- OCT 2024 **UGC Research Travel Grant**, The Hong Kong University of Science and Technology (for attending ECCV'24)
- OCT 2024 **Doctoral Consortium**, ECCV'24, mentored by Prof. Leonidas Guibas
- JUN 2024 **Doctoral Consortium**, IEEE CAI'24, mentored by Prof. Ivor Tsang
- JAN 2024 **2nd Place**, USV-based Obstacle Segmentation Challenge, WACV'24 Workshop on Maritime Computer Vision
- JAN 2023 **ARAP Research Award**, Centre for Frontier AI Research, A*STAR
- JAN 2023 **2nd Place**, USV-based Obstacle Segmentation Challenge, WACV'23 Workshop on Maritime Computer Vision
- OCT 2022 **UGC Research Travel Grant**, The Hong Kong University of Science and Technology (for attending ECCV'22)
- JUN 2022 **Travel Grant**, EPFL CIS Edge AI Summer School 2022
- AUG 2020 **Best Poster Award**, Machine Learning Summer School - Indonesia (MLSS-Indo) 2020
- 2019 - 2024 **Postgraduate Scholarship**, The Hong Kong University of Science and Technology
- JUL 2019 **Scholarship for SENG Summer Camp for Elite Students**, Hong Kong University of Science and Technology
- 2014 - 2015 **Scholarship for Excellent Academic Performance**, HCMIU-VNU

PROFESSIONAL APPOINTMENTS

- 2026 - now **Executive Officer – Secretary**, IEEE Coastal Los Angeles Section, Region 6 (Western US).
- 2025 - now **Postdoctoral Scholar**, The University of California, Los Angeles. **Advisor:** Assoc. Prof. M. Khalid Jawed
Research Projects: Drone Navigation for Biological Pest Control in High Tunnels;
3D Scene Reconstruction of Plants and Farms for Agricultural Robots;
Physics-Guided Latent Space Models for Detecting Occluded Objects;
Collision-Free Path Planning for Robotic Pollination.
- 2025 - now **CEO**, Computer Vision for Developing Countries (NPO). Entity No.: B20250147416, EIN: 39-2483919
- 2023 - 2024 **Researcher**, Centre for Frontier AI Research, A*STAR, Singapore
- 2020 - 2024 **Teaching Assistant**, The Hong Kong University of Science and Technology (HKUST)
Courses: Introduction to Computer Science (COMP 1021); Programming with C++ (COMP 2011);
Multimedia Computing (COMP 4431); Introduction to 3D Design (ISDN 2300);
Physical Prototyping (ISDN 2400); Advanced Digital Design (ISDN 5300 / CSIT 6000L / CSIT 5940).
- 2015 - 2018 **Research Assistant and Teaching Assistant**, HCMIU-VNU
Courses: Introduction to Artificial Intelligence; Principles of Programming Languages;
Digital Image Processing; Theoretical Models for Computing.

PROFESSIONAL ACTIVITIES

- 2026 - NOW **Guest Editor:** IJCV CV4Animals ([Website](#))
Session Co-Chair: AAAI'26
Area Chair: ICASSP'26
- 2024 - NOW **Workshop Organizer:**
Computer Vision for Developing Countries (CV4DC) Workshop (ACCV'24, ICCV'25, ACCV'26, [Website](#))
CV4Animals: Computer Vision for Animal Behavior Tracking and Modeling Workshop (CVPR'26, [Website](#))
- 2021 - NOW **Conference Reviewer:** CVPR'22-26, ICCV'23-25, ECCV'22-26, NeurIPS'24-26, ICML'25, ICLR'25-26, AAAI'25-27, IJCAI'25-26, AISTATS'25-26, IJCNN'27, ICRA'25-26, CoRL'26, WACV'22-27, ACM MM'25-26, ECAI'25, BMVC'26, ACCV'22-24, ICMI'25-26, ISVC'25, ICME'23
- 2021 - NOW **Journal Reviewer:** Springer International Journal of Computer Vision, Springer Artificial Intelligence Review, ACM Transactions on Graphics, IEEE Transactions on Image Processing, IEEE Transactions on Multimedia, IEEE Transactions on Big Data, IEEE Robotics and Automation Letters, IEEE Transactions on Artificial Intelligence, Elsevier Expert Systems With Applications, Elsevier Neurocomputing, Elsevier Computer-Aided Design, Elsevier Engineering Applications of Artificial Intelligence, IET Image Processing, Springer Visual Intelligence, Springer Scientific Reports, PeerJ Computer Science, Vietnam Journal of Computer Science

SELECTED PUBLICATIONS

* equal contribution, † guidance role for juniors, # corresponding author. For a complete list of my publications, please refer to [Google Scholar](#).

- 2026 Tuan-An To, Wong Yuk Kwan, **Tuan-Anh Vu**, Zheng Ziqiang, Sai-Kit Yeung, “MarineEVT: Advancing Event-Centric Marine Video Understanding via Visual Tool Reasoning”, ECCV’26. ([Link](#))
- 2026 Hai Nguyen-Truong, **Tuan-Anh Vu**, and Dang Huynh, “Off-Manifold Refinement: Guiding Video Generators with a Frozen World Model”, BMVC’26. ([Link](#))
- 2026 Trung Thanh Nguyen, **Tuan-Anh Vu**, Duc Viet Le, Yasutomo Kawanishi, Takahiro Komamizu, Ichiro Ide, and Teja Kattenborn, “ForestMamba: Sparse Mamba with Geometry-guided Queries for 3D Forest Point Cloud Segmentation”, BMVC’26. ([Link](#))
- 2026 **Tuan-Anh Vu**, Duc-Thanh Nguyen#, Qing Guo#, Nhat Chung, Binh-Son Hua, Ivor W. Tsang, and Sai-Kit Yeung, “Catch Me If You Can Describe Me: Open-Vocabulary Camouflaged Instance Segmentation with Diffusion”, *International Journal of Computer Vision (IJCV)* and ECCV’24 CV4Ecology Workshop. ([Link](#))
- 2026 **Tuan-Anh Vu**, Ziqiang Zheng, Chengyang Song, Qing Guo, Ivor W. Tsang, and Sai-Kit Yeung, “CamoVid60K: A Large-Scale Video Dataset for Moving Camouflaged Animals Understanding”, *International Journal of Computer Vision (IJCV)* and CVPR’25 CV4Animal (**Oral with Travel Award**). ([Link](#))
- 2026 **Tuan-Anh Vu**, Hai Nguyen-Truong, Ziqiang Zheng, Binh-Son Hua, Qing Guo, Ivor W. Tsang, and Sai-Kit Yeung, “Power of Boundary and Reflection: Semantic Transparent Object Segmentation using Pyramid Vision Transformer with Transparent Cues”, **WACV’26** and ECCV’24 Transparent & Reflective Objects In the wild Workshop. ([Link](#))
- 2026 Gysam Chang, **Tuan-Anh Vu**†, Vivek Alumootil, Harris Song, Deanna Pham, Sangpil Kim#, and M. Khalid Jawed#, “Reconstruction Using the Invisible: Intuition from NIR and Metadata for Enhanced 3D Gaussian Splatting”, **AAAI’26** and CVPR’25 - 2nd Workshop on Neural Fields Beyond Conventional Cameras. ([Link](#))
- 2026 Bohao Qu, Xiaofeng Cao, Bing Li, Menglin Zhang, **Tuan-Anh Vu**, Di Lin, and Qing Guo, “Exploiting Geometric Structures for Modeling Multi-Agent Behaviors: A New Thinking”, **AAAI’26**. **Oral** - 4.3% of 23,680 submissions. ([Link](#))
- 2025 Hai Nguyen-Truong*, E-Ro Nguyen*, **Tuan-Anh Vu**#, Binh-Son Hua, Minh-Triet Tran, and Sai-Kit Yeung, “Vision-Aware Text Features in Referring Image Segmentation: From Object to Context Understanding”, **WACV’25**. ([Link](#))
- 2024 Ziqiang Zheng, Yiwei Chen, Huimin Zeng, **Tuan-Anh Vu**, Binh-Son Hua, Sai-Kit Yeung, “MarineInst: A Foundation Model for Marine Image Analysis with Instance Visual Description”, **ECCV’24**. **Oral** - 2.3% of 8,585 submissions ([Link](#))
- 2024 Yingshu Chen, Huajian Huang#, **Tuan-Anh Vu**, Ka-Chun Shum, Sai-Kit Yeung, “StyleCity: Large-Scale 3D Urban Scenes Stylization”, **ECCV’24**, ECCV’24 CV4Metaverse workshop **Oral** and US Patent (Filling). ([Link](#))
- 2024 **Tuan-Anh Vu***, Srinjay Sarkar*, Zhiyuan Zhang, Binh-Son Hua, and Sai-Kit Yeung, “Test-Time Augmentation for 3D Point Cloud Classification and Segmentation”, **3DV’24**. ([Link](#))
- 2023 Tan-Sang Ha*, Hai Nguyen-Truong*, **Tuan-Anh Vu**†, and Sai-Kit Yeung, “MarineVRS: Marine Video Retrieval System with Explainability via Semantic Understanding”, **OCEANS’23**, Limerick. **Oral** ([Link](#))
- 2023 Quang-Trung Truong, **Tuan-Anh Vu**†, Tan-Sang Ha, Jakub Lokoc, Ajay Joneja, and Sai-Kit Yeung, “Marine Video Kit: A New Marine Video Dataset for Content-based Analysis and Retrieval”, **MMM’23**. **Oral** ([Link](#))
- 2022 **Tuan-Anh Vu**, Duc-Thanh Nguyen, Binh-Son Hua, Quang-Hieu Pham, and Sai-Kit Yeung, “RFNet-4D: Joint Object Reconstruction and Flow Estimation from 4D Point Clouds”, **ECCV’22**. **Oral** - 2.7% of 5,803 submissions ([Link](#))
- 2022 Yingshu Chen, **Tuan-Anh Vu**, Binh-Son Hua and Sai-Kit Yeung, “Time-of-Day Neural Style Transfer for Architectural Photographs”, **IEEE ICCP’22**. **Oral** ([Link](#))

PREPRINTS

- 2026 Hoang M. Truong, Hai Nguyen-Truong, and **Tuan-Anh Vu**#, “EgoPoseProbe: Diagnosing Scene Grounding and Feature Adaptation in Egocentric 3D Body Pose Estimation”, **AAAI’27** (under review). ([Link](#))
- 2026 Trung Thanh Nguyen, Hai Nguyen-Truong, Tu Vo, Hoang M. Truong, and **Tuan-Anh Vu**#, “Cross4D-JEPA: Dense Cross-modal Correspondence Distillation for 4D Point Cloud Representation Learning”, **WACV’27** (under review). ([Link](#))
- 2026 **Tuan-Anh Vu***, Yashas Shashidhara*, M. Khalid Jawed, “VLA-Flight: Lightweight Onboard Vision-Language-Action with Direct IMU Fusion for UAV Control”, **CoRL’26** (under review). ([Link](#))
- 2026 Huyen-Thu Vu, Thanh-Hoang Nguyen-Vo, **Tuan-Anh Vu**#, and Binh P. Nguyen#, “Do Simplicial, Hypergraph, and Spectral Biases Compose in Molecular GNNs? A Controlled Negative Result”, **TMLR** (under review). ([Link](#))
- 2026 **Tuan-Anh Vu**, Xiaoyang Zhao, M. Khalid Jawed, “Information-Theoretic Point Selection and Observability-Constrained EKF for Degeneracy-Aware RGB-D Visual-Inertial Odometry”, **IEEE T-RO** (under review). ([Link](#))
- 2026 **Tuan-Anh Vu**, M. Khalid Jawed, “Physics-Regularized Latent Representations for Hidden Object Inference from Thermal Dynamics”, **IJCV** (under review). ([Link](#))
- 2026 **Tuan-Anh Vu***, Angela Yang*, Mohammad Sadoughi, Amit Kachroo, M. Khalid Jawed, “Consistency and Stability of LLM Reasoning Loops: A Latent Dynamical Systems Perspective”, **NeurIPS’26** (under review). ([Link](#))

- 2026 Alan Nadelsticher Ruvalcaba, Joohyun Lee, Wenying Wu, Benjamin Ostrower, Junling Zhuang, Amir J. Bidhendi, Janie Huang, Luke DeVivo, **Tuan-Anh Vu**[†], Supratik Mukhopadhyay, “EvoReg: Versatile and Robust Point Cloud Registration via Multi-Stage Alignment”, NeurIPS’26 (under review). ([Link](#))
- 2026 Nhat Chung*, Toan Nguyen*, Yun Xing, **Tuan-Anh Vu**, Bohao Qu, Yedi Zhang, Yue Cao, Jie Zhang, Yue Wang, Cheston Tan, Harold Soh, Qing Guo#, Jin Song Dong#, David Hsu#, Daniel Seita#, and Ivor Tsang#, “Towards Human-Centered Safety in Vision-Language-Action Policies for Robotics: A Survey and Outlook”, IEEE T-RL (under review). ([Link](#))
- 2026 Tan-Sang Ha, **Tuan-Anh Vu**#, and Sai-Kit Yeung, “AquaVision: An Edge-Deployed 360° Underwater Panoramic Monitoring System”, MobiCom’26 (under review). ([Link](#))
- 2026 **Tuan-Anh Vu**, Evelyn Zhu, Xiaoyang Zhao, Angela Yang, Akshat Pandya, M. Khalid Jawed, “AgriDrone: A Multi-Sensor Dataset and Lightweight Vision Backbone for Perception in Greenhouse Drone Navigation”, Preprint. ([Link](#))
- 2026 Jaehwan Jeong, Evelyn Zhu, Jinying Lin, Emmanuel Jaimes, **Tuan-Anh Vu**, Jungseock Joo, Sangpil Kim#, M. Khalid Jawed#, “Your VLA Model Already Has Attention Heads For Path Deviation Detection”, Preprint. ([Link](#))
- 2025 Jaehwan Jeong*, **Tuan-Anh Vu***, Radha Lahoti, Vivek Alumootil, Sangpil Kim#, M. Khalid Jawed#, “Vision-Guided Targeted Grasping and Vibration for Robotic Pollination in Controlled Environments”, Preprint. ([Link](#))
- 2025 Vivek Alumootil, **Tuan-Anh Vu**#, “DePT3R: Joint Dense Point Tracking and 3D Reconstruction of Dynamic Scenes in a Single Forward Pass”, Preprint. ([Link](#))
- 2025 Harris Song*, **Tuan-Anh Vu***[†], Sanjith Menon, Sriram Narasimhan, M. Khalid Jawed, “HiddenObject: Modality-Agnostic Fusion for Multimodal Hidden Object Detection”, Preprint. ([Link](#))
- 2025 Jaehwan Jeong, **Tuan-Anh Vu**[†], Mohammad Jony, Shahab Ahmad, Md Mukhlesur Rahman, Sangpil Kim#, M. Khalid Jawed#, “AgriChrono: A Multi-modal Dataset Capturing Crop Growth and Lighting Variability with a Field Robot”, Preprint. ([Link](#))
- 2025 **Tuan-Anh Vu**, Duc-Thanh Nguyen, Binh-Son Hua, Quang-Hieu Pham, and Sai-Kit Yeung, “Reconstruction of Time-Varying Point Clouds via Bidirectional Weakly-Supervised Learning of Spatio-Temporal Cross Features”, Preprint. ([Link](#))
- 2024 Nhat Minh Chung, Sensen Gao, **Tuan-Anh Vu**, Jie Zhang, Aishan Liu, Yun Lin, Jin Song Dong, Qing Guo, “Towards Transferable Attacks Against Vision-LLMs in Autonomous Driving with Typography”, Preprint. ([Link](#))
- 2024 Ziqiang Zheng, Jipeng Zhang, **Tuan-Anh Vu**, Shizhe Diao, and Sai-Kit Yeung, “MarineGPT: Unlocking Secrets of “Ocean” to the Public”, Preprint. ([Link](#))
- 2023 Ziqiang Zheng, Yiwei Chen, Jipeng Zhang, **Tuan-Anh Vu**, Huimin Zeng, Yue W. Tim, and Sai-Kit Yeung, “Exploring Boundary of GPT-4V on Marine Analysis: A Preliminary Case Study”, Preprint. ([Link](#))

REFERENCES

Prof. M. Khalid Jawed

Department of Mechanical and Aerospace Engineering
University of California, Los Angeles

Prof. Duc Thanh Nguyen

School of Information Technology
Deakin University, Australia

Prof. Binh-Son Hua

School of Computer Science and Statistics
Trinity College Dublin, Ireland

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Director of A*STAR Centre for Frontier AI Research (CFAR)
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